

NYC Employment & 311 Service Request Analysis

Comprehensive analysis of New York City employment trends by major industry sector using NYS Department of Labor CES data, and 311 service request patterns from NYC OpenData, covering noise complaints and illegal parking incidents.

Data Sources: NYS DOL CES & NYC OpenData 311

Prepared by Z.ai

April 2026

Table of Contents

Question 1: New York State Department of Labor Employment Statistics Analysis	2
1a. Major Industry Changes Over the Prior Year	2
1b. Comparison of 1-Year vs. 5-Year Industry Changes	4
Question 2: NYC 311 Service Requests Analysis	7
2a. API Query Construction and Data Export	7
2b. Complaint Counts by Agency and Complaint Type	7
2c. Data Visualization and Policy Implications	9
2d. Research Question, Additional Data, and Equation	11

Question 1: New York State Department of Labor Employment Statistics Analysis

1a. Major Industry Changes Over the Prior Year

Using the New York State Department of Labor's Current Employment Statistics (CES) data for the latest available month (February 2026), this analysis examines employment changes across ten major industry supersectors (2-digit NAICS level) in New York City. The data was sourced from the CES CSV file available at <https://dol.ny.gov/statistics-ceszip>, which contains monthly employment figures for New York State, metropolitan areas, and counties from 1990 to the present. The latest month with complete data across all sectors is February 2026, and the 1-year comparison is computed as the difference between February 2026 and February 2025 employment levels. These supersectors represent the broadest industry classifications in the NAICS hierarchy and provide a comprehensive overview of structural shifts in the city's labor market.

The following table presents the February employment levels for 2025 and 2026, along with the absolute change and percentage change for each major industry sector. Industries are sorted by the magnitude of their percentage change, from the largest decline to the largest gain.

Industry (NAICS Supersector)	Feb 2025	Feb 2026	1-Yr Change	1-Yr % Change
Manufacturing (31-33)	53,300	50,200	-3,100	-5.82%
Mining, Logging & Construction (21, 23)	137,000	130,400	-6,600	-4.82%
Other Services (81)	176,800	171,900	-4,900	-2.77%
Leisure & Hospitality (71, 72)	439,500	429,500	-10,000	-2.28%
Private Education & Health (61, 62)	1,318,800	1,297,400	-21,400	-1.62%
Trade, Transportation & Utilities (22, 42-49)	580,400	573,300	-7,100	-1.22%
Professional & Business Services (54-56)	795,300	788,800	-6,500	-0.82%
Information (51)	217,100	221,300	+4,200	+1.93%
Financial Activities (52, 53)	510,400	516,900	+6,500	+1.27%
Government (92)	601,800	611,300	+9,500	+1.58%

Table 1. NYC Major Industry Employment Change, February 2025 to February 2026 (Source: NYS DOL CES)

Industries Experiencing the Greatest Declines: The three industries with the largest percentage declines over the past year are Manufacturing (-5.82%), Mining, Logging and Construction (-4.82%), and Other Services (-2.77%).

Manufacturing has been in a long-term structural decline in New York City, driven by automation, offshoring of production to lower-cost regions, and the city's ongoing transition toward a service-oriented economy. The apparel manufacturing subsector, once a dominant employer, has been particularly hard hit as fashion production has moved overseas. The construction decline is likely tied to higher interest rates throughout 2025 that suppressed new housing starts and commercial development, along with the completion of several major infrastructure projects that had previously buoyed employment in this sector. Other Services, which includes repair services, personal care services, and religious and civic organizations, may be experiencing pressure from inflation reducing consumer discretionary spending on non-essential services.

Industries Experiencing the Greatest Growth: On the positive side, Information (+1.93%), Financial Activities (+1.27%), and Government (+1.58%) showed modest growth. The Information sector's recovery is noteworthy given its volatility in recent years; after peaking in early 2023 and declining through 2024-2025 due to layoffs in tech and media streaming, the sector appears to be stabilizing with growth in areas like AI-related services and digital content creation. Financial Activities continues its steady expansion, reflecting New York City's enduring role as a global financial center, with gains in securities, asset management, and insurance. Government employment growth is largely driven by local government hiring, particularly in education, as the city continues to fill positions that were left vacant during the pandemic-era budget constraints.

Notable Observation: Private Education and Health Services experienced the largest absolute decline (-21,400 jobs), despite being one of the largest sectors. This is particularly significant given that this sector had been the primary engine of job growth over the prior five years (gaining over 260,000 jobs since 2021). The decline may reflect post-pandemic normalization as the surge in healthcare demand subsides, combined with funding pressures on educational institutions and potential staffing shortages leading to unfilled positions rather than layoffs. Leisure and Hospitality also saw a notable decline of 10,000 jobs, signaling that the post-pandemic recovery in tourism, dining, and entertainment may have peaked, with consumers reallocating spending toward essentials amid persistent inflation.

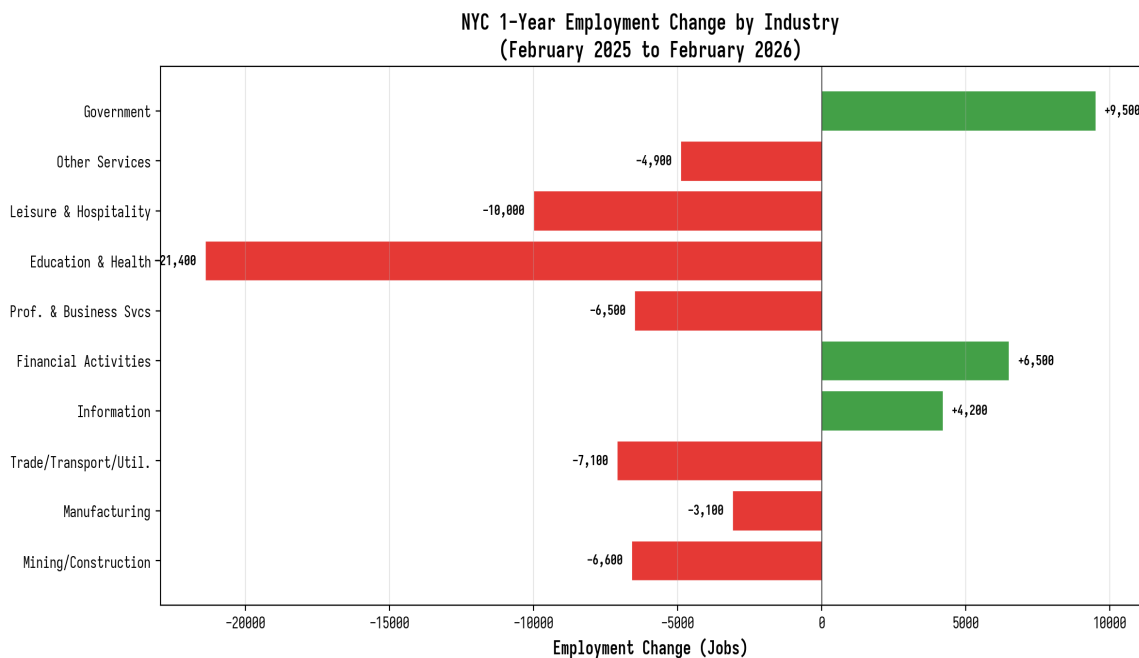


Figure 1. NYC 1-Year Absolute Employment Change by Industry (Feb 2025 to Feb 2026)

1b. Comparison of 1-Year vs. 5-Year Industry Changes

The contrast between the 1-year and 5-year changes reveals dramatic differences in industry trajectories. Over the five-year period from February 2021 to February 2026, the city's overall employment recovered strongly from the COVID-19 pandemic, with Total Nonfarm employment growing by 690,200 jobs (+16.83%), from 4,100,800 to 4,791,000. However, the 1-year change shows a contraction of -39,400 jobs (-0.82%), indicating that the post-pandemic recovery phase has ended and the city is now experiencing a mild employment downturn.

Industry	1-Yr % Change	5-Yr % Change	Difference (5yr - 1yr)
Manufacturing	-5.82%	-3.65%	+2.17 pp
Mining/Construction	-4.82%	-3.91%	+0.91 pp
Other Services	-2.77%	+6.37%	+9.14 pp
Leisure & Hospitality	-2.28%	+85.05%	+87.33 pp
Education & Health	-1.62%	+25.07%	+26.69 pp
Trade/Transport/Utilities	-1.22%	+7.86%	+9.08 pp
Prof. & Business Services	-0.82%	+12.75%	+13.57 pp
Information	+1.93%	+5.23%	+3.30 pp
Financial Activities	+1.27%	+12.13%	+10.86 pp
Government	+1.58%	+5.47%	+3.89 pp
Total Nonfarm	-0.82%	+16.83%	+17.65 pp

Table 2. NYC Major Industry Employment: 1-Year vs. 5-Year Percentage Change (pp = percentage points)

Leisure and Hospitality: The Most Dramatic Divergence. The starkest difference between the 1-year and 5-year trajectories belongs to Leisure and Hospitality. Over five years, this sector grew by an extraordinary +85.05%, adding 197,400 jobs as the city recovered from the devastating pandemic-era losses that had reduced employment to just 232,100 in February 2021. This recovery was fueled by the resurgence of tourism, the reopening of restaurants and bars, and the return of Broadway and live entertainment. However, over the most recent year, the sector contracted by -2.28% (-10,000 jobs), signaling that the rebound has fully matured and possibly overshot sustainable levels. Rising operational costs, labor shortages, and shifting consumer habits (including remote work reducing downtown lunch and happy hour traffic) are now headwinds for this sector.

Private Education and Health Services: From Growth Engine to Contraction. This sector was the largest contributor to 5-year growth (+260,100 jobs, +25.07%), driven by the post-pandemic surge in healthcare demand,

expansion of home health care services, and increased enrollment in private educational institutions. Yet over the past year, it lost 21,400 jobs (-1.62%). This reversal suggests that the healthcare boom is cooling as pandemic-era demand normalizes, and that staffing challenges in nursing homes and residential care facilities may be limiting the sector's ability to maintain employment levels. Additionally, federal and state funding constraints may be pressuring healthcare employers.

Professional and Business Services: Moderating Growth. This sector grew by +12.75% over five years (+89,200 jobs), reflecting the expansion of technology consulting, legal services, and administrative support in the city. However, the 1-year change of -0.82% indicates that this sector has also stalled, likely due to corporate downsizing in professional services, reduced demand for commercial real estate services, and the broader tech industry correction that has affected firms providing IT and consulting services.

Financial Activities: Steady but Slowing. With a 5-year gain of +12.13% and a 1-year gain of +1.27%, Financial Activities remains the most consistently growing private-sector industry. The slowdown from a ~2.4% annualized pace over 5 years to 1.27% in the latest year suggests that even this resilient sector is not immune to broader economic headwinds, including higher interest rates reducing deal volume in securities and investment banking.

Manufacturing and Construction: Persistent Declines. Both sectors show negative changes in both time periods, but the 1-year decline is actually steeper than the 5-year annualized rate for Manufacturing (-5.82% vs. -3.65% total over 5 years), indicating that the structural decline is accelerating. Construction's 1-year decline of -4.82% is worse than its 5-year total of -3.91%, suggesting the sector is now contracting faster than its long-term trend, likely due to the interest rate environment and completion of major capital projects.

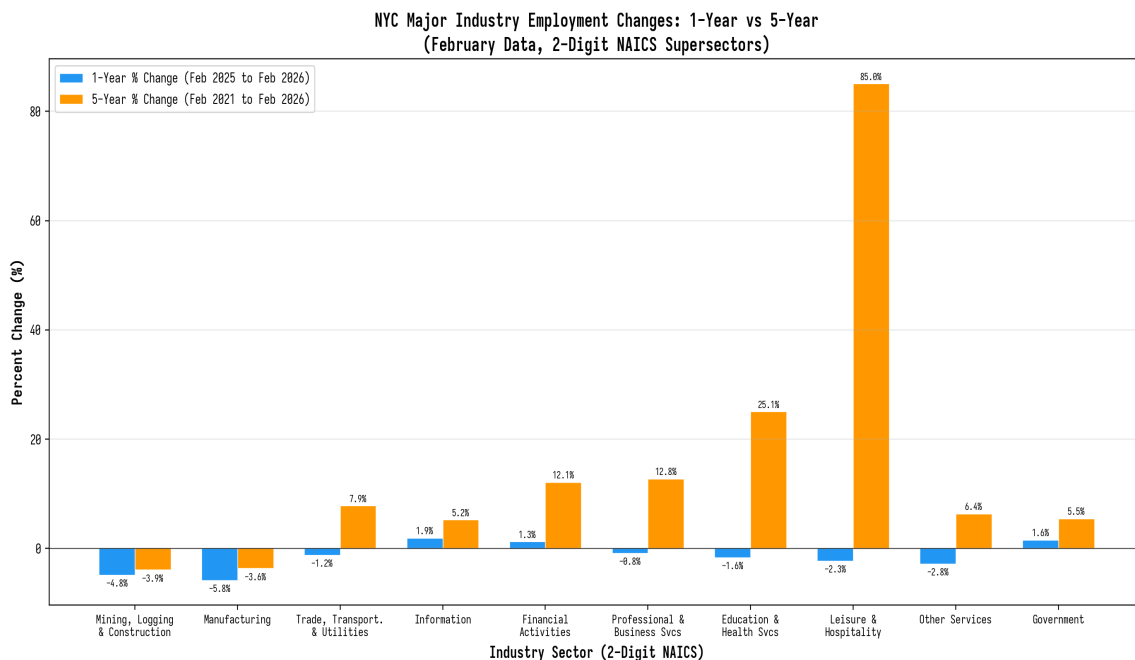


Figure 2. NYC Major Industry Employment Changes: 1-Year vs. 5-Year Comparison

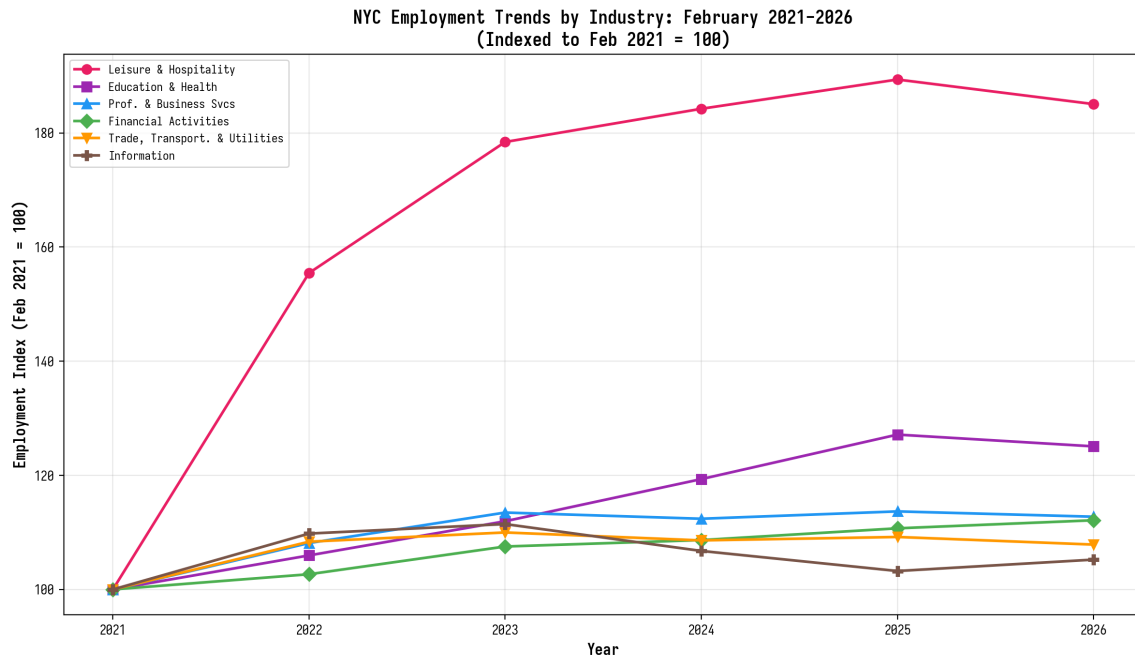


Figure 3. NYC Employment Trends by Industry, February 2021-2026 (Indexed to Feb 2021 = 100)

For additional context, the table below shows the annual average employment levels for each major industry from 2021 to 2026, providing a broader view of the employment trajectory beyond a single month's snapshot. The 2026 figures are based on available data (January-February only) and may shift as the year progresses.

Industry	2021 Avg	2022 Avg	2023 Avg	2024 Avg	2025 Avg	2026 Avg*
Mining/Construction	141,225	143,192	143,575	142,808	137,167	131,750
Manufacturing	54,583	57,775	57,275	54,842	52,517	50,400
Trade/Trans./Util.	551,217	585,825	585,975	581,308	583,292	577,450
Information	220,975	238,442	224,142	224,008	219,867	219,550
Financial Activities	466,050	488,158	502,308	505,908	514,850	515,000
Prof. & Business	722,267	786,517	795,142	796,900	796,250	789,250
Education & Health	1,045,067	1,104,792	1,172,175	1,258,725	1,271,150	1,289,300
Leisure & Hospitality	305,975	402,150	435,617	447,858	451,500	428,750
Other Services	168,008	176,908	177,958	177,417	176,200	171,800
Government	585,842	586,350	592,517	603,967	614,717	609,800

Table 3. NYC Annual Average Employment by Industry, 2021-2026 (*2026 = Jan-Feb average only)

Question 2: NYC 311 Service Requests Analysis

2a. API Query Construction and Data Export

Data was pulled from the NYC OpenData 311 Service Requests dataset using the SODA (Socrata Open Data API) endpoint. The query filters for records where: (1) the created_date falls between December 15, 2021 and March 15, 2022; (2) the agency_name is either "New York City Police Department" or "Department of Housing Preservation and Development"; and (3) the complaint_type starts with "Noise" (capturing all noise subcategories) or equals "Illegal Parking". The query uses SoQL (Socrata Query Language) to filter at the API level, reducing the data transfer volume and ensuring only relevant records are retrieved.

Exact API Query URL:

```
https://data.cityofnewyork.us/resource/erm2-nwe9.json?$where=created_date%20between%20%272021-12-15T00:00:00%27%20and%20%272022-03-15T23:59:59%27%20AND%20(agency_name%3D%27New%20York%20City%20Police%20Department%27%20OR%20agency_name%3D%27Department%20of%20Housing%20Preservation%20and%20Development%27)%20AND%20(starts_with(complaint_type%2C%27Noise%27)%20OR%20complaint_type%3D%27Illegal%20Parking%27)&$select=created_date,agency_name,complaint_type,descriptor,borough,latitude,longitude,unique_key&$limit=50000&$order=unique_key
```

The query was executed with pagination (offset parameter incremented by 50,000) to retrieve all matching records. A total of 198,158 records were retrieved across 4 paginated API calls. The data was exported and saved locally for analysis.

Important Finding: While the query included both the New York City Police Department (NYPD) and the Department of Housing Preservation and Development (HPD) as agency filters, zero records from HPD appeared in the results. This is because HPD handles housing quality complaints (e.g., HEAT/HOT WATER, UNSANITARY CONDITION, PLUMBING, PAINT/PLASTER) rather than noise or illegal parking complaints. An independent verification query confirmed that HPD had 0 records matching the noise or illegal parking complaint types in this date range. HPD's top complaint types during this period were HEAT/HOT WATER (121,072 complaints), UNSANITARY CONDITION (22,267), and PLUMBING (15,738). This finding highlights that the intersection of {NYPD, HPD} and {Noise, Illegal Parking} effectively reduces to {NYPD} alone, which is consistent with each agency's jurisdictional responsibilities.

2b. Complaint Counts by Agency and Complaint Type

The 198,158 complaints in the dataset are exclusively from the New York City Police Department. The complaints are distributed across seven noise subcategories and one illegal parking category. The table below presents the complete breakdown by agency and complaint type, showing that Illegal Parking and Noise - Residential together account for approximately 81.8% of all complaints.

Agency	Complaint Type	Count	% of Total
NYPD	Illegal Parking	86,493	43.6%
NYPD	Noise - Residential	75,543	38.1%

NYPD	Noise - Street/Sidewalk	12,743	6.4%
NYPD	Noise - Vehicle	11,372	5.7%
NYPD	Noise - Commercial	11,114	5.6%
NYPD	Noise - Park	498	0.3%
NYPD	Noise - House of Worship	395	0.2%
Total		198,158	100.0%

Table 4. 311 Complaint Counts by Agency and Complaint Type (Dec 15, 2021 - Mar 15, 2022)

Aggregating the noise subcategories, the overall distribution is as follows: Illegal Parking accounts for 86,493 complaints (43.6%), while all Noise complaints combined total 111,665 complaints (56.4%). Among the noise subcategories, Noise - Residential dominates with 75,543 complaints, representing 67.7% of all noise complaints. This is followed by Noise - Street/Sidewalk (12,743; 11.4%), Noise - Vehicle (11,372; 10.2%), and Noise - Commercial (11,114; 10.0%). Noise from Parks and Houses of Worship are relatively rare, together comprising only 0.8% of noise complaints.

The monthly breakdown reveals a clear seasonal pattern, with complaint volumes higher in January and February 2022 compared to the partial months of December 2021 and March 2022. This likely reflects the higher density of indoor activity during winter months, which increases both residential noise complaints and illegal parking incidents (potentially related to holiday shopping and winter weather conditions affecting parking availability).

Month	Illegal Parking	Noise (All Types)	Total	% of Grand Total
Dec 2021 (partial)	14,619	19,265	33,884	17.1%
Jan 2022	27,859	32,265	60,124	30.3%
Feb 2022	27,166	40,968	68,134	34.4%
Mar 2022 (partial)	16,849	19,167	36,016	18.2%

Table 5. Monthly Breakdown of 311 Complaints by Category

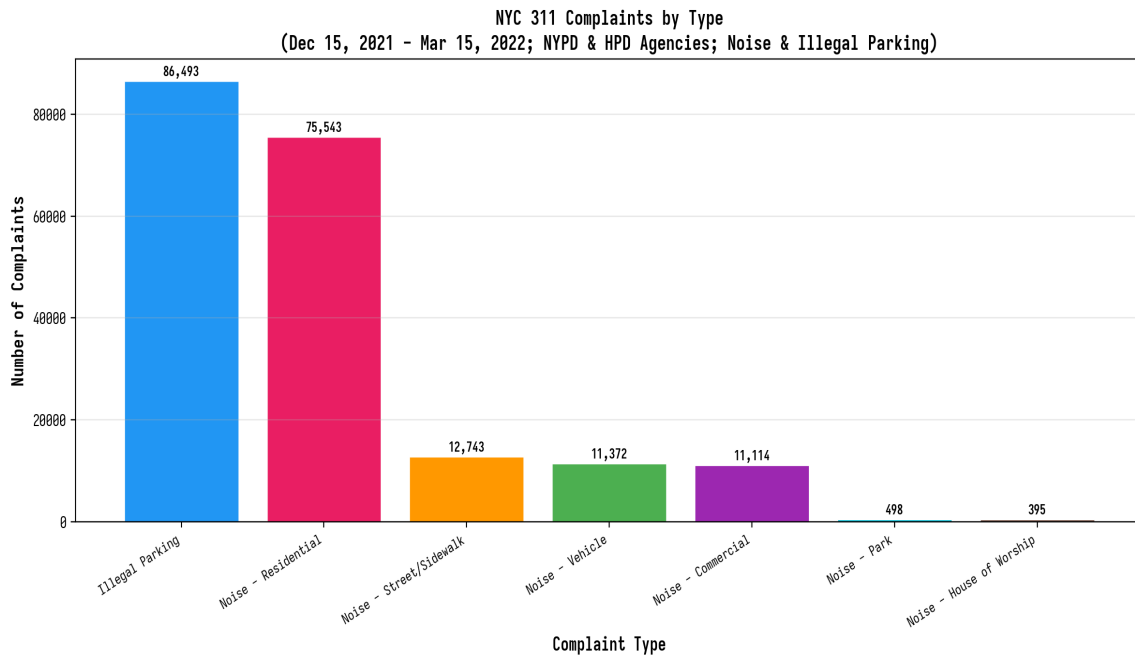


Figure 4. Distribution of 311 Complaints by Type (Dec 2021 - Mar 2022)

2c. Data Visualization and Policy Implications

Recommended Visualization: Grouped Bar Chart of Complaints by Category Over Time. The most effective way to visualize this data is a grouped bar chart showing the monthly count of Illegal Parking complaints versus Noise complaints (all subcategories combined) over the December 2021 to March 2022 period. This visualization highlights both the relative magnitude of each complaint category and how they change over time. The grouped bar chart is preferred over a simple bar chart because it enables direct comparison between the two major complaint categories within each month, while also revealing temporal trends.

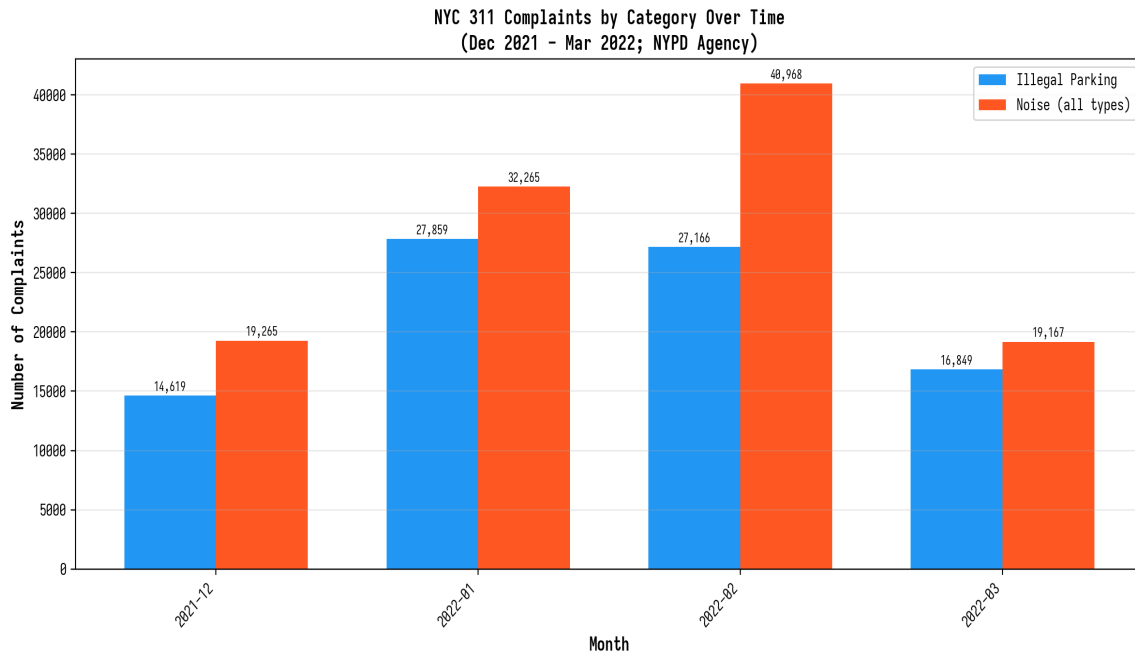


Figure 5. Monthly Trend of 311 Complaints: Illegal Parking vs. Noise (Dec 2021 - Mar 2022)

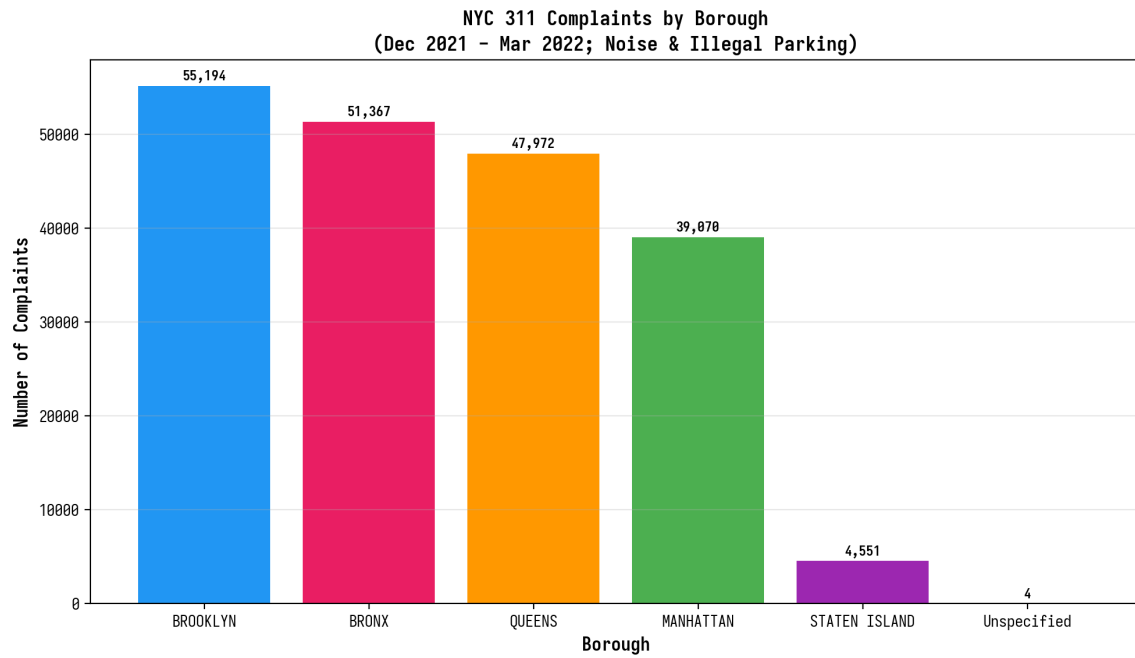


Figure 6. Distribution of 311 Noise and Illegal Parking Complaints by Borough

Borough	Complaint Count	% of Total	Est. Population (2021)	Complaints per 1,000 Residents
Brooklyn	55,194	27.9%	2,576,771	21.4
Bronx	51,367	25.9%	1,424,103	36.1

Queens	47,972	24.2%	2,270,976	21.1
Manhattan	39,070	19.7%	1,601,271	24.4
Staten Island	4,551	2.3%	475,596	9.6

Table 6. Complaint Distribution by Borough with Per-Capita Rates (Population: NYC DOF 2021 Estimates)

What Policymakers Should Know: Several key findings emerge from this analysis that should inform policy decisions. First, the Bronx has a dramatically higher per-capita complaint rate (36.1 per 1,000 residents) compared to the citywide average of approximately 23.5 per 1,000 residents, and more than 3.5 times the rate of Staten Island (9.6). This disparity suggests that the Bronx may be underserved in terms of noise enforcement and parking infrastructure, or that residential density and building conditions in the Bronx generate more conflict between neighbors. The concentration of residential noise complaints (38.1% of all complaints) indicates that quality-of-life issues in residential buildings are a significant concern, particularly in denser boroughs.

Second, illegal parking complaints constitute 43.6% of all complaints, suggesting that parking enforcement is a major resource drain on NYPD. Policymakers should consider whether alternative enforcement mechanisms (such as automated parking enforcement cameras or civilian parking enforcement officers) could reduce the burden on police resources while maintaining compliance. Third, the seasonal pattern shows that noise complaints peak in February, possibly because residents spend more time indoors during cold weather, increasing the likelihood of noise conflicts. This suggests that proactive community mediation programs and building management outreach should be intensified during winter months. Finally, the complete absence of HPD from these results highlights a potential gap in the 311 system: while HPD handles heating and maintenance complaints that may themselves be caused by noise (e.g., faulty boilers, noisy pipes), these are classified differently, and the inter-agency coordination between NYPD and HPD on noise issues that span both agencies' jurisdictions may need strengthening.

2d. Research Question, Additional Data, and Equation

Research Question: How does the density of 311 noise and illegal parking complaints relate to the prevalence of residential building code violations, and what does this relationship reveal about the concentration of urban quality-of-life stressors in disadvantaged neighborhoods?

Additional Data for Merging: To answer this question, I would merge the 311 dataset with the following additional data sources:

1. HPD Housing Maintenance Code Violations Dataset (available via NYC OpenData): This dataset contains records of building code violations issued by the Department of Housing Preservation and Development, including violation types (heat/hot water, plumbing, pests, lead paint, etc.), severity classes (Class A through Class C), and building identifiers. Merging on the building address or BBL (Borough-Block-Lot) number would allow us to link noise complaints at specific addresses with the condition of those buildings.

- American Community Survey (ACS) Demographic Data at the census tract level: Including median household income, poverty rate, racial/ethnic composition, renter-occupied housing percentage, and population density. This would enable us to control for socioeconomic factors and test whether the complaint-violation relationship varies by neighborhood demographics.
- PLUTO (Primary Land Use Tax Lot Output) dataset: Including building age, number of units, lot area, and building class. This would help distinguish whether noise complaints are driven by building conditions (old, poorly insulated buildings) or by neighborhood characteristics.

Simple Equation: The relationship between noise/parking complaints and building violations, controlling for neighborhood characteristics, can be expressed as a regression model:

$$C_i = a + b_1V_i + b_2D_i + b_3B_i + e_i$$

Term	Definition	Measurement
C_i	Complaint density in area i	311 complaints per 1,000 residents per month
V_i	Building violation density in area i	HPD Class B+C violations per 100 residential units
D_i	Demographic disadvantage index for area i	Composite index (poverty rate + renter % + minority %)
B_i	Building characteristics in area i	Mean building age + mean units per building + occupancy rate
a	Intercept	Baseline complaint rate when all predictors are zero
b_1, b_2, b_3	Regression coefficients	Effect size of each predictor on complaint density
e_i	Error term	Unobserved factors affecting complaint density

Table 7. Regression Model Variable Definitions

Why This Relationship Matters for Policymaking: Studying this relationship is critical for several reasons. First, if b_1 is positive and statistically significant, it would indicate that neighborhoods with more building code violations also experience higher rates of noise and parking complaints, suggesting that physical deterioration and quality-of-life nuisance are concentrated in the same areas. This concentration effect means that certain neighborhoods bear a disproportionate burden of urban stressors, which has equity implications for resource allocation.

Second, understanding whether complaints are driven primarily by building conditions (b_1), neighborhood demographics (b_2), or both, has direct implications for intervention strategy. If building conditions are the primary driver, then code enforcement and building investment programs (such as HPD's Alternative Enforcement Program) could simultaneously reduce both violations and complaints. If demographic factors dominate, then community-based mediation programs, cultural competency training for enforcement officers, and investments in affordable housing quality would be more effective.

Third, the coefficient b_2 on the demographic disadvantage index would reveal whether complaints are systematically higher in disadvantaged communities, even after controlling for building conditions. If so, this could indicate either that disadvantaged communities face genuinely worse conditions (more noise, more parking scarcity) or that complaint-filing behavior varies by community. Either interpretation has policy consequences: the former suggests the need for targeted infrastructure investment, while the latter suggests the need for outreach to ensure all communities can effectively access 311 services. Finally, combining 311 data with building violation data creates a more holistic quality-of-life index that policymakers can use to prioritize interventions, allocate enforcement resources, and measure the impact of housing policies on neighborhood well-being over time.